

28 April 2026

Selecting & Sorting

1. Retrieve all columns from the employees table.

SELECT *

FROM employees;



Employees

id	First name	Last name	department	Salary	hire_date	City
1	Alice	Green	IT	70 000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45 000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65 000	2018-07-19	Durban
4	David	Mokoena	Marketing	50 000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72 000	2017-09-30	Johannesburg

2. Find all unique departments.

SELECT DISTINCT department

FROM employees;



department
IT
HR
Finance
Marketing

3. Retrieve first and last names ordered by salary descending.

```
SELECT first_name,  
       last_name
```

```
FROM employees
```

```
ORDER BY salary DESC;
```

↓

first_name	last_name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

4. Retrieve the top 3 highest-paid employees.

```
SELECT id,  
       first_name,  
       last_name,  
       salary
```

```
FROM employees
```

```
ORDER BY salary DESC
```

```
LIMIT 3;
```

↓

id	first_name	last_name	salary
5	Eva	Naidoo	72000
1	Alice	Green	70000
3	Cathy	Zulu	65000

5. Find employees in the IT department.

```
SELECT id,  
       first_name,  
       last_name,  
       department  
FROM employees  
WHERE department = 'IT';
```

↓

id	first_name	last_name	department
1	Alice	Green	IT
5	Eva	Nardoo	IT

6. Find employees in Finance with salary > 60000.

```
SELECT id,  
       first_name,  
       last_name,  
       department,  
       salary  
FROM employees  
WHERE department = 'Finance'  
AND salary > 60000;
```

↓

id	first_name	last_name	department	salary
3	Cathy	Zulu	Finance	65000

7. Find employees in HR or Marketing.

```
SELECT id,  
       first_name,  
       last_name,  
       department  
FROM employees  
WHERE department = 'HR'  
OR department = 'Marketing';
```

↓

id	first_name	last_name	department
2	Brian	Lee	HR
4	David	Mukherjee	Marketing

8. Find employees not in IT.

```
SELECT id,  
       first_name,  
       last_name,  
       department  
FROM employees  
WHERE department <> 'IT';
```

↓

id	first_name	last_name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mukherjee	Marketing

9. Find employees in IT, HR, or Finance using IN.

```
SELECT id,  
       first_name,  
       last_name,  
       department  
FROM employees  
WHERE department IN ('IT', 'HR', 'Finance');
```

↓

id	first_name	last_name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Euly	Finance
5	Eva	Naidu	IT

10. Find employees in IT with salary > 65000 and city Johannesburg.

```
SELECT id,  
       first_name,  
       last_name,  
       department,  
       salary,  
       city  
FROM employees  
WHERE department = 'IT'  
AND salary > 65000  
AND city = 'Johannesburg';
```

↓

id	first_name	last_name	department	salary	city
1	Alice	Green	IT	70000	Johannesburg
5	Eva	Naidu	IT	72000	Johannesburg